

Student Answer Script View



058-END SEMSTER-ODD-2025

Answer Sheet

Course: Computer Science and Engineering**Year/Sem:** Semester 7**36.00****Subject Name:** Deep Learning**50.00****Exam Date:** 27-Nov-2025

Comments submitted by student.

Q.No : 1 B)

Score : 4.50 / 5.00

Analyze the differences between classification and regression in supervised learning. How do the techniques used in each differ in terms of model evaluation?

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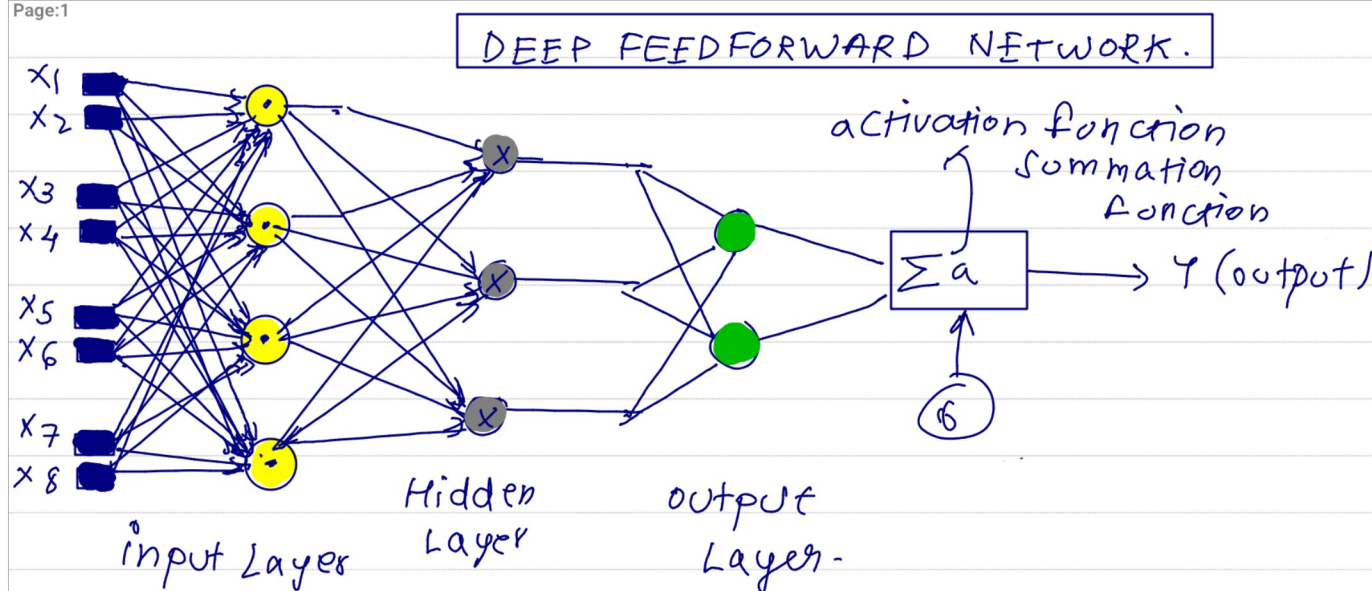
	Classification in supervised learning		Regression in supervised learning
i)	<u>Definition</u> : Classification in supervised learning, where data can be classified into multiple classes by learning from the outside environment	i)	<u>Definition</u> : Linear data is predicted by observing the outside environment, where data points are mapped to become a linear straight line graph, which can help to predict future data points.
ii)	For Example: Dataset of cats and dogs can be classified using supervised learning classification	ii)	For example:- stocks, prediction, steady acceleration of cars, uniform speed can be predicted using regression
iii)	<u>Model Evaluation</u> : model can include DNNs (CNNs, RNNs, FNNs) which are harder to train, high computational complexity.	iii)	<u>Model Evaluation</u> :- include ANN, perceptron training, Easier to train and are simple. computational complexity is low and are generally simple.
iv)	used in classification of labeled data. (clusters)	iv)	used in prediction of linear data

Q.No : 2 A)

Score : 5.00 / 5.00

Analyze the components of a deep feedforward network. How do the input, hidden, and output layers work together to learn from data?

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- Input Layer**:- In our given analogy, we can see we are looking at a deep feedforward network
- i) Input Layer in a neural network is responsible for data fetching and feeding the artificial neurons with data sets ($x_i: x_1, x_2, x_3, x_4, \dots$)
 - ii) data is preprocessed and passed to compute in the hidden layer of the FFN.

Hidden Layer:- For the first layer of the training input data is used, thereafter the preprocessing after the input layer. Hidden layer uses the output from the inputs of input layer to map the data by generalizing it.

- i) In this way, a successful generalization process, the FFN, learns about the data and then upon successful training passes on to the output

Output Layer:- This carries the results post training and results is obtained in this layer.

Q.No : 2 B)

Score : 4.50 / 5.00

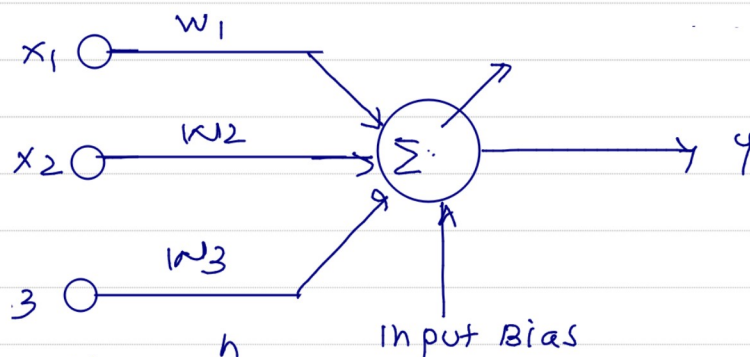
Apply the backpropagation algorithm to adjust weights in a neural network. How does the gradient of the loss function guide the training process?

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Solution:- Backpropagation algorithm is applied to adjust weights in a neural network

i) Let us consider a neural network



$$y = \sum_{i=1}^n w_i x_i$$

$$y = w_1 x_1 + w_2 x_2 + w_3 x_b$$

ii) As the neural network is trained with epoch = 10, it significantly start with a higher loss function

iii) slowly, slowly weights w_i are updated as the neural network progresses learning and loss function limits to minima

iv) Gradient ∇ when reaches minima, we have a 1

Q.No : 3 B)

Score : 4.00 / 5.00

Apply regularization techniques such as L1 and L2 to prevent overfitting. How do they affect the complexity of the model and its performance?

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Solution (i) L1 regularization is a regularization technique, where the unneeded weights in a set are directly set to zero (0) to lower noise which prevents overfitting.

ii) L2 regularization technique, the weights that aren't needed aren't set directly to zero but their values are lowered in order to prevent overfitting.

iii) L1 and L2 regularization play a key role in affecting the complexity of the model and performance.

iv) In case of L1 regularization, weights values are set to 0 if it simplifies the complexity but at the same time, affects the model's performance in a negative manner as weights are directly set to zero, hence losing noise, which is responsible for model learning.

v) In the case of L2 regularization, weights are not completely set to zero, but are reduced, which means that the model's complexity increases but without compromising with the model's performance.

Q.No : 4 A)

Score : 4.50 / 5.00

Evaluate the importance of choosing an appropriate learning rate for neural network optimization. How does the learning rate schedule affect model training and convergence?

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Solution:- It is important to choose an appropriate learning rate for neural network optimization, it does indeed affect model training and convergence in a positive manner.

ii) Let us consider a neural network like DNN when we have a neural network like DNN (Deep Neural Networks), it takes a good solid amount of time to train them (could be days, or weeks), if we do not use a appropriate learning rate.

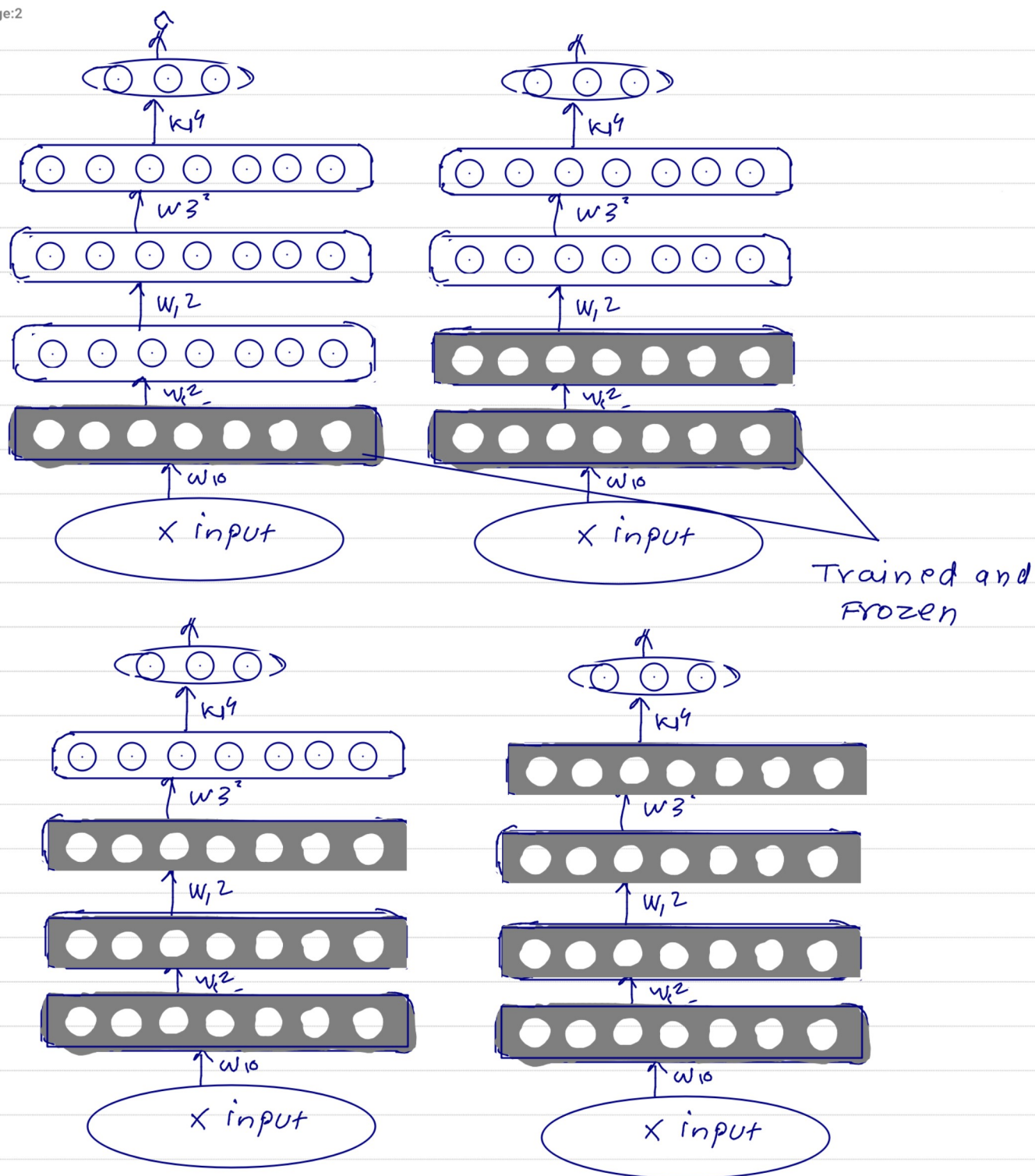
iii) To solve this Greedy Layerwise optimization technique was introduced.

iv) In this technique, we first train the first layer using the input data, and once trained we freeze it

v) once the first layer is completely trained and frozen, we take the output of this to feed the second layer and train it. once the second layer is frozen, we move on to the next layer.

vi) we repeat this process until all layers are trained one by one, and fine tune the model once completed.

vii) this significantly optimizes the learning rate and model training



GREEDY LAYERWISE OPTIMIZATION

Q.No : 4 B)

Score : 4.50 / 5.00

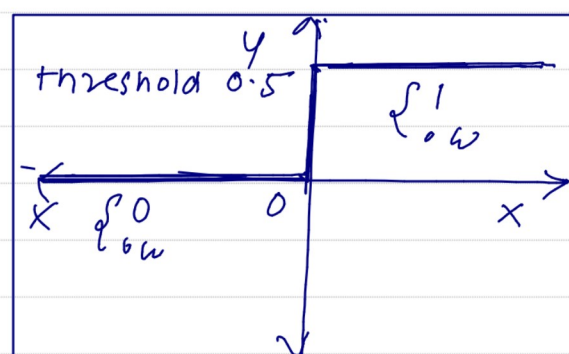
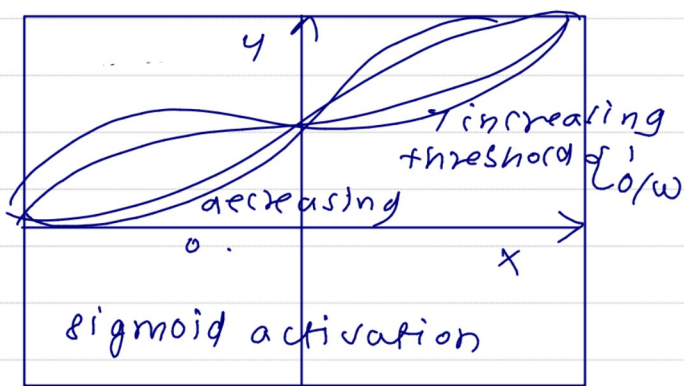
Evaluate the role of activation functions in CNNs. How do they impact the learning ability of the network during training?

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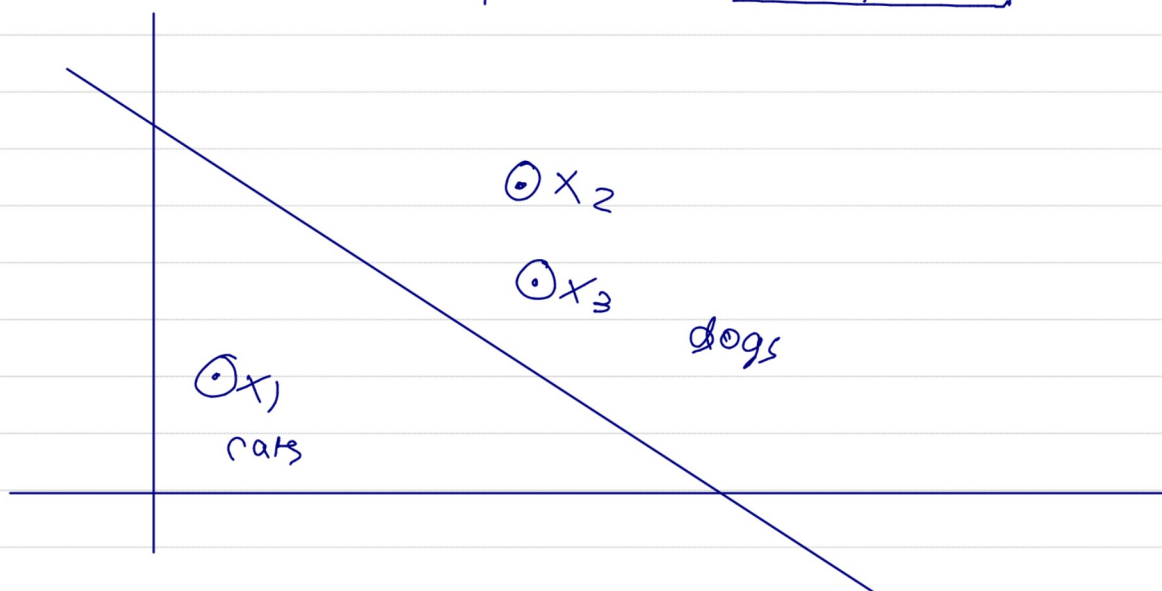
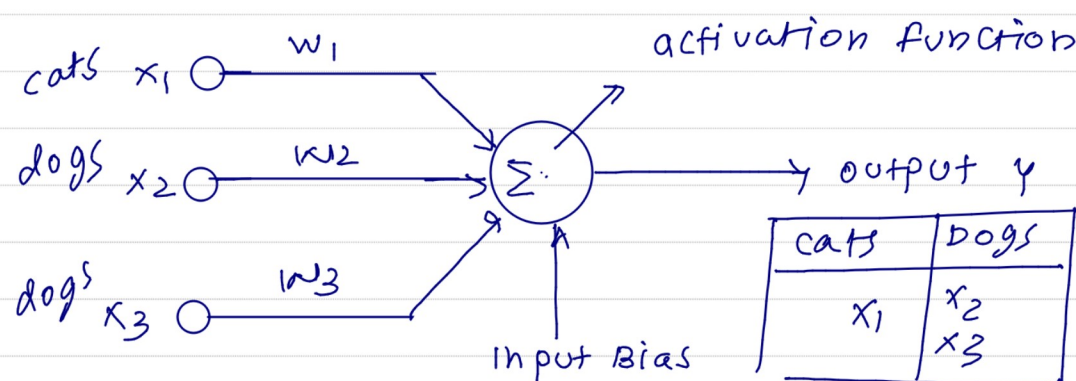
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Solution

- i) Activation functions play a key role in CNNs (Convolutional Neural Networks)
- ii) CNNs are specialized form of neural networks that specialize in image processing, and image classification/prediction.
- iii) Whenever a Image ($32 \times 32 \times 1$) is taken as an input data (x) in a convolutional neural network it passes as it is then convoluted with a Filter (Kernel(k)) to process the image.
- iv) Once processed into filters, an activation function threshold $\begin{cases} 0 \\ 1 \end{cases}$ (ReLU, sigmoid, tanh) then helps the image in binary classification for a successful output.
- v) Activation functions can be of many types like sigmoid, tanh, ReLU, we can use them depending on the use case.



- vi) For example, if we want to classify the images of cats and dogs image dataset using CNNs, Activation function will help in the segregation of the two different datasets.



Q.No : 5 A)

Score : 4.50 / 5.00

Evaluate the impact of vanishing and exploding gradients in RNNs. How do these issues affect the network's ability to learn long-term patterns?

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Solution:-

Vanishing Gradients: As training in a network is executed with time (t), gradients in a network start to get smaller and smaller in size eventually small enough to vanish in the training of RNN.

- i) This significantly impacts the RNNs, as there remains very less data, where the model can generalize on
- ii) This significantly affects the neural network's (RNNs) ability to learn long-term patterns and can be fixed by adding more noise in the dataset.

Exploding Gradients: (i) As the training of the neural network progresses, with time (t), gradients in the network, gain larger and larger in size, large enough, which stops the training process.

- i) This significantly impacts the neural network's because the gradient size, so large that the model starts memorizing patterns instead of generalizing it.
- ii) This causes the training to overfit thus affecting ability to learn long-term patterns.

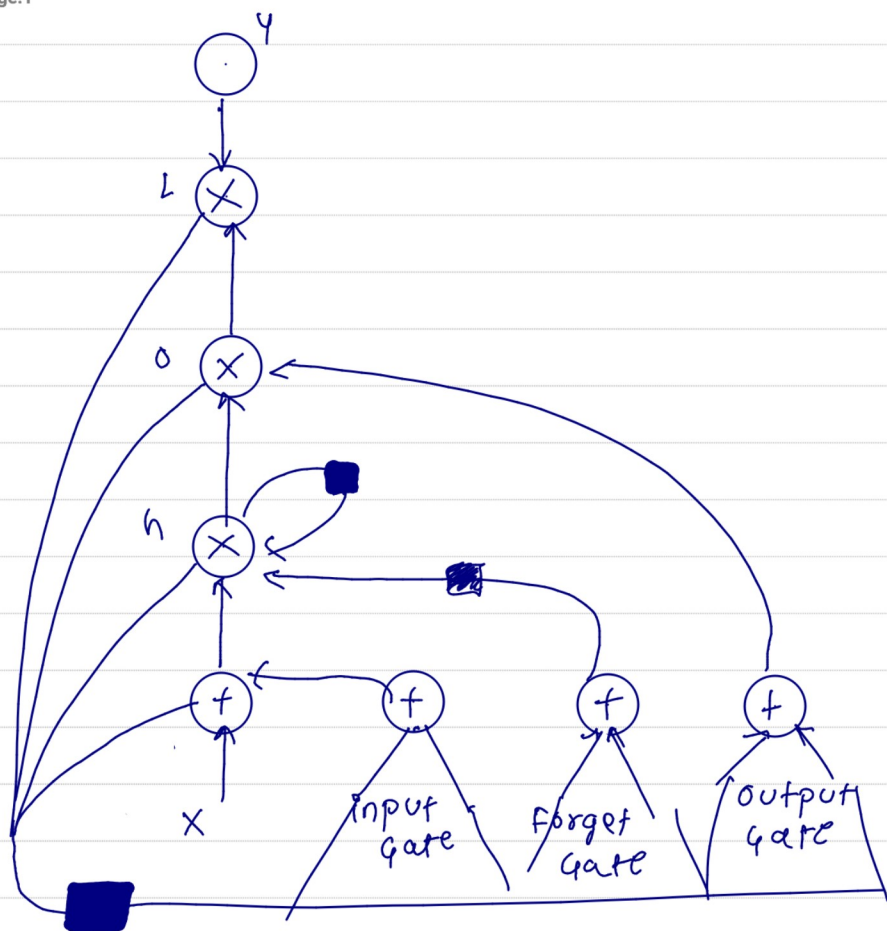
Q.No : 5 B)

Score : 4.50 / 5.00

Evaluate the significance of the input, forget, and output gates in LSTMs. How do they contribute to the network's ability to remember long-term dependencies?

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LSTM (Long short term Memory)

Input Gate : i) The input gate in the LSTM is responsible for feeding the input in the LSTM network.

- ii) Input information is passed in the LSTM's body, whenever a significant information is required by the input x
- iii) The input gates play a key importance in feeding the network $\otimes x$ whenever there is a necessity of data.

Forget gate : (i) Forget gate is responsible to remove existing information or (forget) existing information

in the LSTMS network.

- ii) Whenever, LSTMS operate to find recurring patterns in RNN, when the information that is stored in the hidden layer is no-longer required, Forget gate allows the deletion of the information in the LSTMS.

Output Gate: Output Gate is responsible for feeding output to the network.

- ii) LSTMS, are designed in such a way that they contain self-loops for better architectures.
- iii) The output gate feeds the computed output to a network, which helps the network's ability to remember long-term dependencies.